Parktown Boys' High School

Geography Department

Grade 12 Map Work Assessment

MEMO

Examiner: Ms A Clayton
Moderator: Mr A Meintjes

Time: 1 Hour
Total: 60 Marks
Date: March 2026

QUESTION 1 – MULTIPLE CHOICE (10 marks)

1.1 The questions below are based on the 1: 50 000 topographical map 2531BC MALELANE as well as the orthophoto map 2531 BC 16 RIVERSIDE. Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number on your answer booklet.

1.1 **D**
1.2 **A**
1.3 **C**
1.4 **A**
1.5 **B**
1.6 **A**
1.7 **A**
1.8 **C**
1.9 **D**
1.10 **A**

(10 x 1) (10)

QUESTION 2 – CALCULATIONS (20 marks)

2.1	Give the true bearing from the railway station in Malelane in block E1 to spot height 522 in block E4. **098 to 101° (AC answer 99/100)** (1x2) (2)
2.2	Using your answer in QUESTION 2.1, use the information on the topographical map and information below to calculate the **current** magnetic bearing from the railway station in Malelane in block E1 to spot height 522 in block E4. Show ALL calculations. Mean magnetic declination 18°28' West of True North (January 2014). Mean annual change 5' Westwards. $Current\ MB = TB + Current\ MD$ **Difference in years: 2026-2014 = 12 years (1)** **Total change: 12 years x 5' = 60' (1)** **Present magnetic declination: 18°28' + 60' = 19°28' (1) West of True North (1)** **Magnetic Bearing = 098 to 101 + 19°28' = 117 to 120°19°28'** (1x5) (5)

2.3.	Calculate the area covered by orthophoto on the topographical map. Give the answer in km². Show ALL calculations. **Correct measurement of each side in cm: 3.7/3.8 cm (length), 3.3/3.4cm (breadth) (1)** **Correct conversions to km: 1.85/1.9km x 1.65/1.7km (1)** **Correct answer in km²: 2.48 to 3.23 km² (no unit, -1 mark) (2)** (4 x 1) (4)
2.4	What is the difference in height between the contour line at 7 and the spot height at 8 on the orthophoto map? **382 - 295 = 87m (Range: 82m - 90m)(1)** (1x1) (1)
2.5	Use the answer to QUESTION 2.4 to calculate the gradient from the contour line at 7 to spot height 8. Show ALL calculations Use the map distance of 4,5 cm. **VI = 82m - 90m (from 2.4)** **HE= 4,5 cm x 100m = 450 m (1)** **Gradient = 82-90 / 450 (1) Correct substitution** **= 1 : 5,49 (Range: 5 to 5,49) (1)** (3x1) (3)
2.6	Refer to the cross-section from the orthophoto map from 7 in block C4 to spot height 328 in block E5 below and answer the questions that follow.
	1.6.1. If you stand at spot height 382, will you be able to see the person at spot height 326? Give a reason for your answer. **Yes (1)** **There are no obstacles or peaks (1)** (1x2) (2)
	1.6.2. Calculate the vertical exaggeration of the cross-section shown above. Show ALL calculations. **Correct conversion of VS: 1cm = 20m = 2000cm (1)** **Correct use of formula: 1:2000 / 1:10 000 (1)** **Answer: 5 times (must have times or x) (1)** (1x3) (3)

QUESTION 3 – MAP AND PHOTO INTERPRETATION (20 marks)

3.1	Refer to the Umgwena river in the topographical map.
	3.1.1. In which direct is the river flowing to? **North East** (1x1) (1)
	3.1.2. Provide reason to your answer in QUESTION 3.1.1. **Angle at which tributaries join the main river (2)** **Spot heights / contour lines decrease towards the north east (2)** **[ANY ONE]** (1x2) (2)
3.2	3.2.1. A wind that blows at night from spot height 405 in block B3 in a westerly direction to the Umgwenya River, is known as a/an... wind. **C** (1x1) (1)
	3.2.2. Identify the negative environmental impact that the wind identified in QUESTION 3.2.1 can have on the Leopard Creek Golf Estate in block B2. **Frost can kill plants/vegetation (2)** (1x2) (2)
3.3	Refer to the annual climate graph for Malelane below and answer the questions that follow. **Note:** Temperature is depicted by the line graph and rainfall is depicted by the bar graph.
	3.3.1. Malelane does not receive rainfall all year long. In which season does Malelane receive most of its rainfall? **Summer** (1x1) (1)
	3.3.2. Using the topographical map, give one piece of evidence of the area not receiving rainfall all year. **Prescence of non-perennial rivers/streams;** **Reservoirs to store water.** **ANY ONE** (1x2) (2)
3.4	In winter, pollution levels are increased due to the inversion layer cause by the subsidence of the Kalahari anticyclone over the region.
	3.4.1. Using the topographical map, identify the main cause of the air pollution in the area. **Sugar refinery;** **Also accept (vehicle emissions from main road N4, agricultural practice of cultivating crops)** (1x1) (1)

	3.4.2. Explain why the inversion layer causes pollution to be worse in winter in the area. **Cold, dense air is trapped near the ground.** **The inversion layer prevents vertical air movement.** **There is little convection.** **No uplift → no dispersion of pollutants** **ANY TWO** (2x2) (4)
3.5	Refer to block E1and E2 on the orthophoto map.
	3.5.1. Which time of the day (morning/midday/afternoon) was the photograph taken? **Midday** (1x1) (1)
	3.5.2. Give a reason for your answer to QUESTION 3.5.1. **No shadows present** **Sun directly overhead** **ANY ONE** (1x2) (2)
3.6	Refer to the orthophoto map. Is the feature at 9 a golf course or a dam? **Dam** (1x1) (1)
3.7	Give a reason for the choice of location for the golf course. **Flatter area better for gold course** **Water from river for grass ifrrigation** **Scenic beauty of the river** **ANY ONE** (1x2) (2)

QUESTION 4 – GEOGRAPHIC INFORMATION SYSTEMS (10 marks)

4.1	Refer to area covered by blocks B3 and C3 on the topographical map.
	4.1.1. Define the concept data layering. **Combining /Putting different type of information together.** (1x1) (1)
	4.1.2. Name one data layers with examples found in block **D5** and block **E5.** **Infrastructure: Road & Built-up area** **Drainage: River** **Vegetation: Rec & Golf course** **Topography: Relief , Contour lines** (1x2) (2)
4.2	Refer to diagram **A** and gridblocks 1 and 2 below.
	4.2.1. Explain the concept spatial resolution? **The amount of details and clarity of an image.** (1x2) (2)
	4.2.2. The spatial resolution of image in Diagram A is **high**. Match the gridblocks at 1 **or** gridblocks at 2 with the resolution represented by diagram. Support your answer with a reason **Diagram A : Gridblock 1** **Reason: It is represented in many smaller pixels or blocks** (1 +2) (3)
4.3	Malelane is frequently impacted by tropical cyclones in summer. Suggest ONE method on how the Malelane municipality can use Remote Sensing to raise awareness to the public on an approaching tropical cyclone. **The municipality can use satellite images to track the path and development of the tropical cyclone and make them available to the public so they are well aware.** (1x2) (2)